

HP Financial — Technical Design Document (TDD)

Document	Technical Design Document
Product	HP Financial — CA Practice Management Software
Version	1.0 (Phase 1)
Date	2026-06-25
Companion	PRD.md (requirements). This TDD is the engineering spec.
Status	Draft for review

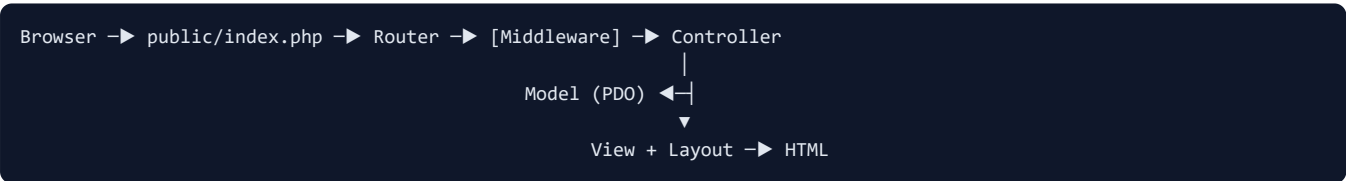
This document turns the PRD into a build-ready engineering specification: architecture, database DDL, routing, class design, shared subsystems, validation, security, deployment, and the screen inventory that drives the design phase. **No development begins until this TDD and the screen designs are approved.**

1. Architecture & Request Lifecycle

Pattern: hand-rolled **MVC** on a **single front controller**. No framework.

Stack: PHP 8.1+, MySQL/MariaDB (PDO), Bootstrap 5, vanilla JS + light jQuery.

Request lifecycle: 1. All web traffic hits `public/index.php` (via `.htaccess` rewrite). 2. `index.php` bootstraps: load `.env` → `config/config.php` → autoloader → start `Session` → build `Request`. 3. Router matches `METHOD` + `path` against `config/routes.php` to a `Controller@method` plus a middleware list. 4. **Middleware** runs in order (`AuthMiddleware` , then `RoleMiddleware` with the route's required permission). On failure → redirect to login / 403. 5. The **Controller** validates input (`Validator` , `Csrf`), calls **Models** (PDO prepared statements), and renders a **View** within a layout. 6. View output is escaped via `e()` ; flash messages and the auth user are available to all views.



2. Tech Stack & Dependencies

Concern	Choice
Language	PHP 8.1+
DB	MySQL 8 / MariaDB 10.4+, via PDO (prepared statements only)
Front-end	Bootstrap 5 (local), vanilla JS, minimal jQuery
Config	<code>vlucas/phpdotenv</code> (read <code>.env</code>)
Encryption	<code>sodium_*</code> (libsodium, bundled in PHP 8.1) for portal credentials
Autoload	Composer PSR-4 (<code>App\</code> → <code>app/</code>)

Removed from Phase 1: PHPMailer / any mailer — Phase 1 is **in-app notifications only** (no email/SMS/WhatsApp).

`composer.json` (Phase 1): `vlucas/phpdotenv` only.

3. Directory Structure

```
hp-financial/
├── public/                # web root (only this is exposed)
│   ├── index.php         # front controller
│   ├── .htaccess         # rewrite to index.php; deny dotfiles
│   └── assets/           # css, js, img, vendor (bootstrap)
├── app/
│   ├── Core/             # Router, Database, Controller, Model, Auth,
│   │                       # Session, Csrf, Validator, Request, Crypto, View
│   ├── Middleware/       # AuthMiddleware, RoleMiddleware
│   ├── Controllers/      # + Controllers/Services/* (one per service)
│   ├── Models/           # User, Lead, Customer, ServicePayment, *Job ...
│   ├── Views/            # layouts/, auth/, leads/, customers/, services/ ...
│   └── Helpers/functions.php
├── config/               # config.php, routes.php
├── database/             # schema.sql, seed.sql, migrations/
├── storage/              # logs/, uploads/customers/{id}/ (deny web access)
├── cron/                 # reminders.php (daily)
├── tools/                # md2pdf.py
└── .env / .env.example / .gitignore / composer.json
```

4. Configuration & Environment

`.env` (never committed):

```
APP_ENV=production
APP_URL=https://crm.example.com
APP_KEY=base64:...      # 32-byte key for sodium credential encryption
DB_HOST=localhost
DB_NAME=hp_financial
DB_USER=hp_user
DB_PASS=*****
SESSION_NAME=hp_session
REMINDER_LEAD_DAYS=3    # default; also overridable in settings table
UPLOAD_MAX_MB=5
```

`config/config.php` exposes typed constants (e.g. `APP_KEY`, `DB_*`, paths) and fails fast if `APP_KEY` is missing.

5. Routing & Route Map

`config/routes.php` returns an array of routes: `[method, path, handler, middleware[], permission]`. Path params use `{id}`.

Method	Path	Handler	Access (permission)
GET/POST	/login	AuthController@login	guest
POST	/logout	AuthController@logout	auth
GET/POST	/password/change	AuthController@changePassword	auth
GET	/	DashboardController@index	auth
GET	/employees	EmployeeController@index	employees.view
GET/POST	/employees/create	EmployeeController@create	employees.manage
GET/POST	/employees/{id}/edit	EmployeeController@edit	employees.manage
POST	/employees/{id}/status	EmployeeController@toggleStatus	employees.manage
POST	/employees/{id}/reset-password	EmployeeController@resetPassword	employees.manage

Method	Path	Handler	Access (permission)
GET/POST	/roles	RoleController@index	roles.manage
GET	/leads	LeadController@index	leads.view
GET/POST	/leads/create	LeadController@create	leads.create
GET/POST	/leads/{id}	LeadController@show	leads.view
POST	/leads/{id}/activity	LeadController@addActivity	leads.edit
POST	/leads/{id}/assign	LeadController@assign	leads.edit
POST	/leads/{id}/convert	LeadController@convert	customers.create
GET	/customers	CustomerController@index	customers.view
GET/POST	/customers/create	CustomerController@create	customers.create
GET	/customers/{id}	CustomerController@show	customers.view
POST	/customers/{id}/update	CustomerController@update	customers.edit
POST	/customers/{id}/documents	CustomerController@uploadDoc	customers.edit
POST	/customers/{id}/services	CustomerController@assignServices	customers.edit
GET/POST	/customers/{id}/{service}/jobs	Services\{Service}Controller@store	services.edit
GET/POST	/jobs/{service}/{id}/edit	Services\{Service}Controller@edit	services.edit
POST	/jobs/{service}/{id}/payments	PaymentController@store	payments.record
GET	/work-board	ServiceController@board	services.view
GET	/reports	ReportController@index	reports.view
GET	/reports/{type}	ReportController@show	reports.view
GET	/reports/{type}/export	ReportController@export	reports.view
GET	/notifications	NotificationController@index	auth
POST	/notifications/{id}/read	NotificationController@markRead	auth
GET/POST	/settings	SettingController@index	settings.manage
GET/POST	/masters/{type}	ServiceController@masters	masters.manage

{service} ∈ income-tax, gst, accounting, loan-subsidy, mutual-fund, insurance, govt-dept, certificate, deeds-agreement, company-compliance .

6. Core Class Design (app/Core)

Class	Responsibility	Key methods
Router	match request → handler; run middleware	add() , dispatch(Request)
Database	single PDO connection (singleton)	pdo() , instance()
Model	base: CRUD via PDO prepared statements	find() , all() , where() , insert() , update() , delete() , paginate()
Controller	base: render views, redirects, JSON, auth helpers	view() , redirect() , json() , user() , authorize(\$perm)
Auth	login/logout, current user, hashing	attempt(mobile,pwd) , login(user) , logout() , user() , check()

Class	Responsibility	Key methods
Session	safe session wrapper	start(), get/set/forget, flash(), regenerate()
Csrf	token issue + verify	token(), field(), verify(\$t)
Validator	rule-based input validation	make(\$data,\$rules), fails(), errors()
Request	encapsulate input/files/method	input(), file(), method(), only()
Crypto	encrypt/decrypt portal credentials	encrypt(\$plain), decrypt(\$cipher) (sodium secretbox + APP_KEY)
View	render PHP templates into layout	render(\$tpl,\$data,\$layout)

Middleware - AuthMiddleware — redirect to /login if not authenticated; enforce must_change_password → /password/change . - RoleMiddleware — check the route permission against the user's role permissions; 403 page otherwise.

Models — one per table; service job models extend Model and add service-specific helpers (e.g. default due-date computation).

ServicePayment model centralizes the fees engine (\$9.2).

7. Database Schema (DDL)

Engine InnoDB, charset utf8mb4. All tables have id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY and (where noted) timestamps. Fks use ON DELETE RESTRICT except logs/children which CASCADE. **Shared capture-once fields (firm_name, gst_number, pan_number, aadhaar_number) live on customers only** — per-service job tables reference the customer and do not duplicate them.

7.1 Auth, roles, permissions

```
CREATE TABLE roles (
  id      BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name    VARCHAR(50) NOT NULL,
  slug    VARCHAR(50) NOT NULL UNIQUE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE permissions (
  id      BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name    VARCHAR(80) NOT NULL,
  slug    VARCHAR(80) NOT NULL UNIQUE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE role_permissions (
  role_id      BIGINT UNSIGNED NOT NULL,
  permission_id BIGINT UNSIGNED NOT NULL,
  PRIMARY KEY (role_id, permission_id),
  FOREIGN KEY (role_id) REFERENCES roles(id) ON DELETE CASCADE,
  FOREIGN KEY (permission_id) REFERENCES permissions(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE users (
  id              BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name            VARCHAR(120) NOT NULL,
  mobile          VARCHAR(15) NOT NULL UNIQUE, -- login identifier
  email           VARCHAR(150) NULL,
  password_hash   VARCHAR(255) NOT NULL,
  role_id         BIGINT UNSIGNED NOT NULL,
  status          ENUM('active','inactive') NOT NULL DEFAULT 'active',
  must_change_password TINYINT(1) NOT NULL DEFAULT 1,
  last_login_at   DATETIME NULL,
  created_by      BIGINT UNSIGNED NULL,
  created_at      DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at      DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
                  ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (role_id) REFERENCES roles(id),
  INDEX (status)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

7.2 Lead masters & leads

```
CREATE TABLE lead_types (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(80) NOT NULL,
  status ENUM('active','inactive') NOT NULL DEFAULT 'active'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE lead_categories (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(80) NOT NULL,
  status ENUM('active','inactive') NOT NULL DEFAULT 'active'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE leads (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(120) NOT NULL,
  mobile VARCHAR(15) NOT NULL,
  lead_type_id BIGINT UNSIGNED NULL,
  category_id BIGINT UNSIGNED NULL,          -- interested category
  state VARCHAR(80) NULL,
  district VARCHAR(80) NULL,
  tehsil VARCHAR(80) NULL,
  village VARCHAR(80) NULL,
  contact_person VARCHAR(120) NULL,
  follow_up_date DATE NULL,                 -- optional
  status ENUM('new','contacted','qualified','won','lost')
  NOT NULL DEFAULT 'new',
  assigned_to BIGINT UNSIGNED NULL,
  notes TEXT NULL,
  created_by BIGINT UNSIGNED NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
  ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (lead_type_id) REFERENCES lead_types(id),
  FOREIGN KEY (category_id) REFERENCES lead_categories(id),
  FOREIGN KEY (assigned_to) REFERENCES users(id),
  INDEX (status), INDEX (assigned_to), INDEX (follow_up_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE lead_activities (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  lead_id BIGINT UNSIGNED NOT NULL,
  user_id BIGINT UNSIGNED NULL,
  type ENUM('note','call','meeting','status_change') NOT NULL DEFAULT 'note',
  description TEXT NULL,
  follow_up_at DATE NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (lead_id) REFERENCES leads(id) ON DELETE CASCADE,
  INDEX (lead_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;
```

7.3 Customers (shared capture-once fields live here)

```
CREATE TABLE customers (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  lead_id BIGINT UNSIGNED NULL,             -- source lead
  name VARCHAR(120) NOT NULL,
  mobile VARCHAR(15) NOT NULL,
  firm_name VARCHAR(160) NULL,              -- SHARED
  pan_number VARCHAR(10) NULL,              -- SHARED
  aadhaar_number VARCHAR(12) NULL,          -- SHARED
  gst_number VARCHAR(15) NULL,              -- SHARED
  email VARCHAR(150) NULL,
  bank_details VARCHAR(255) NULL,
  state VARCHAR(80) NULL,
  district VARCHAR(80) NULL,
  tehsil VARCHAR(80) NULL,
```

```

village          VARCHAR(80) NULL,
contact_person   VARCHAR(120) NULL,
customer_type    VARCHAR(40) NULL,
status           ENUM('active','inactive') NOT NULL DEFAULT 'active',
assigned_to      BIGINT UNSIGNED NULL,
created_by       BIGINT UNSIGNED NULL,
created_at       DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
updated_at       DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
                  ON UPDATE CURRENT_TIMESTAMP,
FOREIGN KEY (lead_id) REFERENCES leads(id),
FOREIGN KEY (assigned_to) REFERENCES users(id),
INDEX (firm_name), INDEX (gst_number), INDEX (mobile)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE customer_attributes (      -- extensible shared fields
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  `key` VARCHAR(60) NOT NULL,
  `value` VARCHAR(255) NULL,
  UNIQUE KEY uq_cust_key (customer_id, `key`),
  FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE customer_documents (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  doc_type ENUM('pan','aadhaar','bank_passbook','other') NOT NULL,
  text_value VARCHAR(100) NULL,
  file_path VARCHAR(255) NULL,
  uploaded_by BIGINT UNSIGNED NULL,
  uploaded_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
  INDEX (customer_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

7.4 Services master, file statuses, customer_services

```

CREATE TABLE services (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(80) NOT NULL,
  code VARCHAR(40) NOT NULL UNIQUE,      -- income_tax, gst, ...
  status ENUM('active','inactive') NOT NULL DEFAULT 'active'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE file_statuses (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(60) NOT NULL,
  sort_order INT NOT NULL DEFAULT 0,
  status ENUM('active','inactive') NOT NULL DEFAULT 'active'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE customer_services (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  service_id BIGINT UNSIGNED NOT NULL,
  status ENUM('active','inactive') NOT NULL DEFAULT 'active',
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  UNIQUE KEY uq_cust_service (customer_id, service_id),
  FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
  FOREIGN KEY (service_id) REFERENCES services(id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

7.5 Shared fees engine — `service_payments` (polymorphic)

```

CREATE TABLE service_payments (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  job_type VARCHAR(30) NOT NULL,      -- income_tax, gst, accounting, ...

```

```

job_id    BIGINT UNSIGNED NOT NULL,
amount    DECIMAL(12,2) NOT NULL,
payment_mode ENUM('cash','bank') NOT NULL,
received_date DATE NOT NULL,
recorded_by BIGINT UNSIGNED NOT NULL,
recorded_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
remarks VARCHAR(255) NULL,
FOREIGN KEY (recorded_by) REFERENCES users(id),
INDEX (job_type, job_id), INDEX (received_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

7.6 Per-service job tables

Common columns on every *_jobs table: id, customer_id (FK→customers), file_status_id (FK→file_statuses), fees_amount DECIMAL(12,2), comment TEXT, assigned_to (FK→users), created_by, created_at, updated_at. Receipts come from service_payments. **No firm_name / gst_number columns** (read from customer).

```

CREATE TABLE income_tax_jobs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  sub_type ENUM('itr','tds','audit','other') NOT NULL,
  title VARCHAR(120) NULL,          -- 'other' free-text
  form_type ENUM('24Q','26Q') NULL, -- TDS only
  financial_year VARCHAR(9) NULL,    -- e.g. 2025-26
  due_date DATE NULL,
  file_status_id BIGINT UNSIGNED NULL,
  fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
  comment TEXT NULL,
  assigned_to BIGINT UNSIGNED NULL,
  created_by BIGINT UNSIGNED NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
  FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
  INDEX (customer_id), INDEX (due_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE gst_profiles (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  gst_user_id VARCHAR(80) NULL,
  form_name VARCHAR(80) NULL,
  filing_type ENUM('monthly','quarterly') NULL,
  status ENUM('active','inactive') NOT NULL DEFAULT 'active',
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4; -- gst_number is on customers (shared)

CREATE TABLE gst_jobs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  gst_profile_id BIGINT UNSIGNED NULL,
  category ENUM('return','registration','audit') NOT NULL,
  return_type ENUM('gstr1','gstr3b') NULL,          -- returns
  audit_form_type ENUM('9','9_9c') NULL,            -- audit
  trn_number VARCHAR(20) NULL,                      -- registration
  arn_number VARCHAR(20) NULL,                      -- registration
  gst_user_id VARCHAR(80) NULL,
  financial_year VARCHAR(9) NULL,
  period_label VARCHAR(20) NULL,                    -- 'Jun-2026' / 'Apr-Jun 2026'
  due_date DATE NULL,                               -- default 10th/20th
  filing_date DATE NULL,
  file_status_id BIGINT UNSIGNED NULL,
  fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
  comment TEXT NULL,
  assigned_to BIGINT UNSIGNED NULL,
  created_by BIGINT UNSIGNED NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,

```

```

    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (gst_profile_id) REFERENCES gst_profiles(id),
    FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
    INDEX (customer_id), INDEX (category), INDEX (due_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

```

CREATE TABLE accounting_jobs (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    customer_id BIGINT UNSIGNED NOT NULL,
    frequency ENUM('monthly','quarterly','yearly') NOT NULL,
    financial_year VARCHAR(9) NULL,
    period_label VARCHAR(20) NULL,
    due_date DATE NULL, -- default 5th
    file_status_id BIGINT UNSIGNED NULL,
    fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
    comment TEXT NULL,
    assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
    INDEX (customer_id), INDEX (due_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

```

CREATE TABLE loan_subsidy_jobs (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    customer_id BIGINT UNSIGNED NOT NULL,
    sub_type ENUM('udyam','cma_report','loan') NOT NULL,
    bank_name VARCHAR(120) NULL, -- CMA & Loan
    loan_amount DECIMAL(14,2) NULL,
    loan_officer_name VARCHAR(120) NULL,
    loan_officer_number VARCHAR(15) NULL,
    due_date DATE NULL,
    file_status_id BIGINT UNSIGNED NULL,
    fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
    comment TEXT NULL,
    assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
    INDEX (customer_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

```

CREATE TABLE mutual_fund_jobs (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    customer_id BIGINT UNSIGNED NOT NULL,
    account_open TINYINT(1) NOT NULL DEFAULT 0,
    ucc_number VARCHAR(40) NULL,
    due_date DATE NULL,
    file_status_id BIGINT UNSIGNED NULL,
    fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
    comment TEXT NULL,
    assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
    INDEX (customer_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

```

CREATE TABLE mutual_fund_investments (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    job_id BIGINT UNSIGNED NOT NULL,
    investment_type ENUM('sip','lumpsum','transfer') NOT NULL,
    target_amount DECIMAL(14,2) NULL,
    achieved_amount DECIMAL(14,2) NULL,
    FOREIGN KEY (job_id) REFERENCES mutual_fund_jobs(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```



```

CREATE TABLE insurance_jobs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  referred_by VARCHAR(120) NULL,
  company_name VARCHAR(120) NULL,
  policy_type ENUM('fresh','port') NULL,
  status VARCHAR(60) NULL,
  claim_amount DECIMAL(14,2) NULL,      -- claim assist
  follow_up_date DATE NULL,             -- claim assist
  due_date DATE NULL,
  file_status_id BIGINT UNSIGNED NULL,
  fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
  comment TEXT NULL,
  assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
  FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
  INDEX (customer_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE insurance_job_types (      -- multi-select
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  insurance_job_id BIGINT UNSIGNED NOT NULL,
  type VARCHAR(40) NOT NULL,             -- health/term/vehicle/claim_assist/...
  FOREIGN KEY (insurance_job_id) REFERENCES insurance_jobs(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE govt_dept_jobs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  sub_type ENUM('audit','gst','tds') NOT NULL,
  officer_name VARCHAR(120) NULL,
  officer_contact VARCHAR(15) NULL,
  contact_person VARCHAR(120) NULL,
  audit_type VARCHAR(80) NULL,           -- audit
  gst_form_type VARCHAR(20) NULL,        -- gst (GSTR-7/other)
  gst_user_id VARCHAR(80) NULL,
  tds_form_type ENUM('24G','26Q','24R') NULL, -- tds
  tan_number VARCHAR(15) NULL,
  financial_year VARCHAR(9) NULL,
  due_date DATE NULL,
  file_status_id BIGINT UNSIGNED NULL,
  fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
  comment TEXT NULL,
  assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
  FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
  FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
  INDEX (customer_id), INDEX (sub_type)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE govt_dept_credentials (    -- ENCRYPTED values ($9.4)
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  job_id BIGINT UNSIGNED NOT NULL,
  cred_type ENUM('gst_portal','traces','it_portal','ain_24q','ain_26q') NOT NULL,
  username VARBINARY(512) NULL,          -- ciphertext
  password VARBINARY(512) NULL,         -- ciphertext
  FOREIGN KEY (job_id) REFERENCES govt_dept_jobs(id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE certificate_jobs (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  customer_id BIGINT UNSIGNED NOT NULL,
  form_name VARCHAR(120) NULL,
  certificate_type VARCHAR(120) NULL,
  due_date DATE NULL,
  file_status_id BIGINT UNSIGNED NULL,

```

```

    fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
    comment TEXT NULL,
    assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
    INDEX (customer_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE deeds_agreement_jobs (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    customer_id BIGINT UNSIGNED NOT NULL,
    type VARCHAR(120) NULL,
    due_date DATE NULL,
    file_status_id BIGINT UNSIGNED NULL,
    fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
    comment TEXT NULL,
    assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
    INDEX (customer_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE company_compliance_jobs (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    customer_id BIGINT UNSIGNED NOT NULL,
    sub_type ENUM('audit','company_registration','compliances','other') NOT NULL,
    form_name VARCHAR(120) NULL,
    cs_number VARCHAR(40) NULL,          -- company registration
    cs_name VARCHAR(120) NULL,
    cs_contact VARCHAR(15) NULL,
    financial_year VARCHAR(9) NULL,
    due_date DATE NULL,
    file_status_id BIGINT UNSIGNED NULL,
    fees_amount DECIMAL(12,2) NOT NULL DEFAULT 0,
    comment TEXT NULL,
    assigned_to BIGINT UNSIGNED NULL, created_by BIGINT UNSIGNED NULL,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES customers(id) ON DELETE CASCADE,
    FOREIGN KEY (file_status_id) REFERENCES file_statuses(id),
    INDEX (customer_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

7.7 System tables

```

CREATE TABLE notifications (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    user_id BIGINT UNSIGNED NOT NULL,
    type VARCHAR(40) NOT NULL,
    title VARCHAR(160) NOT NULL,
    message VARCHAR(255) NULL,
    link VARCHAR(255) NULL,
    is_read TINYINT(1) NOT NULL DEFAULT 0,
    created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
    INDEX (user_id, is_read)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE reminders (
    id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
    source_type ENUM('lead_followup','service_job') NOT NULL,
    source_ref VARCHAR(40) NULL,          -- service code for service_job
    source_id BIGINT UNSIGNED NOT NULL,
    due_date DATE NOT NULL,
    remind_on DATE NOT NULL,

```

```

channel ENUM('in_app') NOT NULL DEFAULT 'in_app', -- Phase 1: in-app only
status ENUM('pending','sent') NOT NULL DEFAULT 'pending',
sent_at DATETIME NULL,
INDEX (remind_on, status)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE settings (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  `key` VARCHAR(60) NOT NULL UNIQUE,
  `value` VARCHAR(255) NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

CREATE TABLE activity_log (
  id BIGINT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
  user_id BIGINT UNSIGNED NULL,
  action VARCHAR(60) NOT NULL,          -- created/updated/deleted/payment...
  entity VARCHAR(60) NOT NULL,
  entity_id BIGINT UNSIGNED NULL,
  detail VARCHAR(255) NULL,
  created_at DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP,
  INDEX (entity, entity_id), INDEX (user_id)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

7.8 Seed data (`seed.sql`)

- Roles: `admin`, `employee`.
- Permissions: `employees.view/manage`, `roles.manage`, `leads.view/create/edit`, `customers.view/create/edit`, `services.view/edit`, `payments.record`, `reports.view`, `settings.manage`, `masters.manage`. Admin → all; employee → operational subset (view/create/edit leads & customers, services.edit, payments.record, reports.view).
- One admin user (mobile + temp password, `must_change_password=1`).
- Services (10), file_statuses (Pending/In-Progress/Filed/Completed), lead_types (Referral, Social Media), lead_categories (sample).
- Settings: `reminder_lead_days=3`.

8. RBAC & Permission Matrix

Permission	admin	employee
employees.view / manage	✓	✗
roles.manage / settings.manage / masters.manage	✓	✗
leads.view / create / edit	✓	✓
customers.view / create / edit	✓	✓
services.view / edit	✓	✓
payments.record	✓	✓ (own jobs)
reports.view	✓	✓ (scoped to self)

Row-level rule: employees see/edit records **assigned to them**; admins see all. Enforced in model query scopes + controller `authorize()`.

9. Shared Subsystems

9.1 Shared customer fields (capture-once)

`firm_name`, `gst_number`, `pan_number`, `aadhaar_number` are read from `customers` and rendered **read-only** on every service-job form. Editable only on the customer profile. Additional shared fields can be added via `customer_attributes` without schema changes. Service controllers call a `Customer::sharedFields($id)` helper to populate the read-only block.

9.2 Fees / part-payment engine

- A job's `fees_amount` is the agreed fee. Receipts are rows in `service_payments` keyed by (`job_type` , `job_id`).
- `ServicePayment::summary($jobType,$jobId) → [received, balance, count]` where `received = SUM(amount)` , `balance = fees_amount - received` .
- **Write rule:** allowed if `user.role=admin` OR `job.assigned_to=user.id` . Enforced in `PaymentController@store` .
- Every insert/update writes `activity_log ( action='payment' )` .

9.3 File uploads

- Accept `pdf/jpg/jpeg/png` , max `UPLOAD_MAX_MB` . Validate MIME + extension + size. Generate random filename; store at `storage/uploads/customers/{customer_id}/` . Persist relative `file_path` .
- `storage/` is outside web root / denied by `.htaccess` ; downloads are streamed through an authenticated controller action (never a direct URL).

9.4 Credential encryption (Govt Dept portal logins)

- `Crypto::encrypt()` uses `sodium_crypto_secretbox` with a per-record nonce and the 32-byte `APP_KEY` ; ciphertext stored as `VARBINARY` .
- Decrypted only when an authorized user (admin or job assignee) opens the job; shown masked with a reveal toggle. Never logged.

9.5 Notifications & reminders (in-app only)

- `cron/reminders.php` runs **daily** (Hostinger cron). For each open service job with a `due_date` and each lead `follow_up_date` , if `due_date - reminder_lead_days <= today` and no `reminder` exists → insert a `reminder ( channel='in_app' )` and a `notification` for the assignee.
- Overdue jobs (`due_date < today`, not completed) surface on dashboard/work board.

9.6 Activity log

Written on create/update/delete of leads, customers, jobs, employees, and on all payment changes. Viewable by admin.

10. Per-Service Module Design

- Each service has a controller in `app/Controllers/Services/` and a model per `*_jobs` table. A shared `partials/_fees.php` + `partials/_shared_fields.php` render the fees panel and read-only customer fields on every job form.
- **Conditional UI** (progressive disclosure, server-validated):
- *Income Tax* `sub_type` toggles TDS `form_type` / *Other* title.
- *GST* `category` toggles Returns vs Registration (TRN/ARN) vs Audit fields; `return_type` sets due-date default (10th/20th).
- *Govt Dept* `sub_type=tds` reveals credential blocks (TAN→TRACES+IT; AIN-24Q; AIN-26Q) saved to `govt_dept_credentials` (encrypted).
- *Insurance* multi-select types; `claim_assist` reveals claim fields.
- *Mutual Fund* `account_open` reveals UCC; investment-type rows (target/achieved) added dynamically → `mutual_fund_investments` .
- Default due-date helpers compute 10th/20th (GST) and 5th (Accounting) from the selected period, editable by the user.

11. Validation Rules (server-side, `Validator`)

Field	Rule
mobile	required, 10 digits <code>^[6-9]\d{9}\$</code>
password (set)	required, min 8, confirmed
PAN	<code>^[A-Z]{5}[0-9]{4}[A-Z]\$</code> (optional)

Field	Rule
GST number	<code>^\d{2}[A-Z]{5}\d{4}[A-Z]\d[A-Z\d]{2}\$</code> (optional)
Aadhaar	12 digits (optional)
financial_year	<code>^\d{4}-\d{2}\$</code>
amounts	numeric ≥ 0 , 2 decimals
dates	valid date; <code>received_date</code> $\leq$ today
file upload	mime in allowlist, $\leq$ max MB

All POST routes require a valid CSRF token; failures return 419 + flash.

12. UI Structure & Screen Inventory

Layouts: `auth.php` (centered card) and `app.php` (fixed sidebar + topbar + content). **Shared components:** sidebar nav, topbar (search, notifications bell, user menu), page header with actions, filter bar, data table + pagination, status pills, cards/stat tiles, tabs, timeline, modals, toasts, empty states.

Screens (designed in the next phase): Login; Set/Reset password; Admin dashboard; Employee dashboard; Employees list & add/edit; Roles & permissions; Leads list, add/edit, detail (timeline); Lead masters; Customers list; Customer profile (Profile/Documents/Services tabs); Convert-lead; Service job forms (Income Tax, GST, Govt Dept, Insurance, Mutual Fund + shared pattern for the rest); Fees/part-payment modal; Work board; Reports index + report view; Notifications center; Settings & masters.

13. Security Design (checklist)

- Passwords: bcrypt; forced change on first login / after reset.
- Sessions: regenerate on login; httponly+secure cookies; idle timeout.
- CSRF on all writes; output escaping on all echoes; PDO prepared statements.
- RBAC in middleware + row-level scoping; admin-only masters/settings.
- Uploads validated + stored outside web root; authenticated download proxy.
- Portal credentials encrypted at rest (sodium + `APP_KEY`); masked in UI.
- `.htaccess` denies `.env`, `app/`, `storage/`, `config/`; force HTTPS.
- Errors logged to `storage/logs/`; generic error page in production.

14. Deployment (Hostinger)

- PHP 8.1+; create MySQL DB/user (hPanel) $\rightarrow$ `.env`.
- Point domain doc-root to `public/` (preferred) or use the app-outside- `public_html` fallback (adjust `index.php` paths).
- Import `database/schema.sql` then `seed.sql` (phpMyAdmin).
- Generate `APP_KEY`; set `storage/` writable (755), web-denied.
- Add **daily cron**: `php /home/.../cron/reminders.php`.
- Enable free SSL; force HTTPS. Configure DB backups.

15. Coding Conventions

- PSR-4 autoload (`App\`), PSR-12 style, 4-space indent.
- Controllers thin; business rules in models/services; no SQL in controllers.

- One class per file; `PascalCase` classes, `camelCase` methods, `snake_case` DB columns. All queries parameterized. Centralized helpers in `app/Helpers/functions.php` (`e()`, `url()`, `old()`, `flash()`, `can()`).